

Breakdown of the Linearized Radiation-Pressure Description for Non-Gaussian Sensing States

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Abstract

We quantify, in a single-mode model, where the linearized per-frequency description of radiation-pressure backaction (“squeezing + rotation with $\chi(\Omega)$ ”) stops being trustworthy. Comparing the linearized ponderomotive shear against the un-linearized Kerr interaction at matched strength $\chi_{\text{eff}} = 4\chi_K\bar{n}$, we find that for a coherent carrier the linearization is excellent and systematically improvable (error $\propto \chi_{\text{eff}}^2$, falling as $1/\bar{n}$ at fixed χ_{eff}), while non-Gaussian states break it at *first order* in χ_{eff} : a Fock state is $\sim 5\%$ off already at $\chi_{\text{eff}} = 0.05$, and squeezed and cat states reach $\mathcal{O}(10\%)$ by $\chi_{\text{eff}} \approx 0.2$ – 0.4 . The failure is qualitative, not perturbative: the linearized model predicts several dB of ponderomotive anti-squeezing for states on which the exact interaction produces none (Fock) or bends the state into non-Gaussian shapes no linear map can represent (cat, squeezed). We discuss the mechanism — the linearization is a mean-field expansion, and these states have no single mean field — and propose next steps: a two-mode carrier-sideband model to separate artifact from physics, a joint few-frequency-mode simulation to test per-frequency independence, and a homodyne-CFI milestone for honest standard-quantum-limit statements.

1 Two light-only models of backaction

We work in the per-frequency, light-only picture: the mirror is eliminated and radiation pressure acts on a single optical mode as a unitary. All simulations use quadratures $X_\theta = (ae^{-i\theta} + a^\dagger e^{i\theta})/\sqrt{2}$, so $[x, p] = i$ and the vacuum variance is $1/2$; the metrological signal is a quadrature displacement generated by $X_{\theta_{\text{sig}}}$.

Exact (Kerr) form. Tracing out a mirror that responds linearly to the intensity of the light leaves the field with an intensity-dependent phase, i.e. a self-Kerr unitary

$$U_K = \exp(-i\chi_K \hat{n}^2), \quad (1)$$

with χ_K the accumulated ponderomotive phase per photon squared (for an explicit mirror of frequency ω_m and coupling g_0 , $\chi_K = g_0^2(\omega_m t - \sin \omega_m t)/\omega_m^2$; in the free-mass limit $\chi_K = \hbar G^2 \tau^3 / 12m$). This is the polaron/Bose–Jacobs–Knight result and involves no assumption about the state of the light.

Linearized (shear) form. The frequency-domain description used in interferometry (Caves; Kimble *et al.*) linearizes the field about a carrier and retains a Gaussian unitary acting on the fluctuations: the ponderomotive shear

$$U_S = \exp(-i\frac{\chi}{2} X_0^2) : \quad x \mapsto x, \quad p \mapsto p - \chi x, \quad (2)$$

equivalently a rotation–squeeze–rotation with squeeze parameter $r = \text{arcsinh}(\chi/2)$, with $\chi \leftrightarrow \mathcal{K}(\Omega)$ the Kimble factor. Acting on covariance matrices it is the symplectic congruence $C \mapsto SCST^\top$, $S = \begin{pmatrix} 1 & 0 \\ -\chi & 1 \end{pmatrix}$, which is how the two-photon formalism treats *any* input state, Gaussian or not.

2 The bridge between the two models

Expanding Eq. (1) about a coherent carrier, $a = \alpha + \delta a$ with $\bar{n} = \alpha^2$,

$$\hat{n}^2 = \bar{n}^2 + 2\sqrt{2}\bar{n}^{3/2}\delta x + 2\bar{n}(\delta x^2 + \delta\hat{n}) + \mathcal{O}(\delta^3), \quad (3)$$

so the Kerr acts on the fluctuations as a constant phase, a fixed displacement, a rotation, and the shear of Eq. (2) with

$$\boxed{\chi_{\text{eff}} = 4\chi_K \bar{n}} \quad (4)$$

The first neglected term is cubic, $\propto \alpha\delta x\delta\hat{n}$, smaller than the retained quadratic terms by one factor of the fluctuation-to-carrier ratio. The comparison metric is chosen to be insensitive to the rotation and displacement that also appear at quadratic order: we compare the *largest eigenvalue of the output quadrature covariance matrix* — the ponderomotive anti-squeezing magnitude, a rotation-invariant — between (i) the exact Kerr output, simulated in a convergence-checked Fock basis, and (ii) the linearized prediction $\lambda_{\text{max}}(SC_{\text{in}}S^\top)$ computed directly from the input covariance. Note that this covariance-level metric is the *most charitable* comparison for the linearized model, since it is blind to all non-Gaussian structure of the output; the breakdown scales below are therefore lower bounds.

Frequency dependence. To express the comparison as a function of sideband frequency rather than abstract coupling, we evaluate the bridge along the physical frequency dependence of the ponderomotive coupling, the Kimble factor of the tuned interferometer,

$$\chi_{\text{eff}}(\Omega) = \mathcal{K}(\Omega) = \frac{2(I_0/I_{\text{SQL}})\gamma^4}{\Omega^2(\gamma^2 + \Omega^2)}, \quad (5)$$

with γ the cavity half-bandwidth and $\mathcal{K} = 1$ at $\Omega = \gamma$ for $I_0 = I_{\text{SQL}}$. The interpretation is a frequency-multiplexed preparation: each sideband mode carries an identical copy of the state and constitutes its own single-mode channel with strength $\mathcal{K}(\Omega)$. Backaction — and with it the linearization question — is then strong at low frequency and negligible at high frequency.

3 Results

Table 1 and Figs. 1–3 summarize the comparison. Three mechanisms, one per class of state:

- **Coherent: the linearization works and is systematically improvable.** The error grows quadratically in χ_{eff} and falls as $1/\bar{n}$ at fixed χ_{eff} — the classical-carrier limit becoming exact. This is the regime the frequency-domain formalism was built for, and at interferometric photon numbers it is effectively exact.

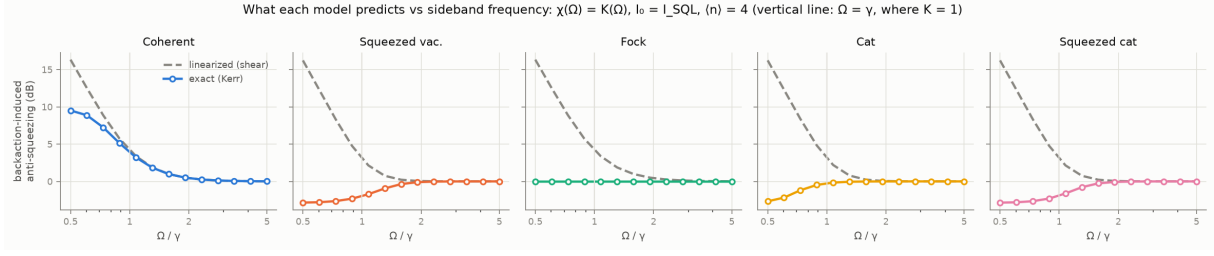


Figure 1: What each model predicts across the band: backaction-induced anti-squeezing of the noisiest quadrature (dB above the input state) versus sideband frequency, with $\chi_{\text{eff}}(\Omega) = \mathcal{K}(\Omega)$ at $I_0 = I_{\text{SQL}}$ [Eq. (5)], for the exact Kerr (solid, colored) and the linearized shear (dashed, gray). Above $\Omega \approx 2\gamma$ backaction is negligible and the models agree trivially; below $\Omega \approx \gamma$ the linearized model predicts a steeply growing noise ($\mathcal{K} \propto \Omega^{-4}$ in this region) that the exact model does not produce for any non-Gaussian family (Fock: exactly none at any frequency). Only the coherent carrier tracks the linearized prediction into the backaction-dominated band.

- **Fock: the linearized model hallucinates backaction.** A number eigenstate has *zero* intensity fluctuations, so there is nothing to fluctuate the radiation-pressure force: $U_K|n\rangle$ is a global phase and the exact model produces no distortion at all. The linearized model, blind to number statistics, predicts full ponderomotive squeezing; the entire tabulated error is that phantom prediction, which is why it appears at first order in χ_{eff} .
- **Cat and squeezed states: no single mean field exists.** The even cat is a superposition of two carriers $\pm\alpha$; each branch would linearize well about its own mean field, but the exact interaction bends the two branches oppositely, and no single linear map represents that. Squeezed vacuum has $\langle a \rangle = 0$ with all of its intensity carried by fluctuations, so the “small parameter” of the expansion is $\mathcal{O}(1)$ by construction; the resulting S-shaped curl of the exact output, versus the straight line the linearized model predicts, is visible across the band in Fig. 2.

Table 1: Relative disagreement $|\lambda_{\text{Kerr}} - \lambda_{\text{shear}}|/\lambda_{\text{shear}}$ at matched strength, Eq. (4). Top: state families at fixed input $\bar{n} = 4$. Bottom: coherent states of increasing $\bar{n}$. Squeezed families are oriented anti-squeezed along the signal generator.

	$\chi_{\text{eff}} = 0.05$	0.2	0.8	1.6
Coherent	2×10^{-5}	0.0010	0.039	0.16
Squeezed vac.	0.0071	0.100	0.59	0.84
Fock	0.049	0.181	0.54	0.77
Cat	0.0028	0.043	0.42	0.76
Squeezed cat	0.0057	0.083	0.58	0.84
Coherent, $\bar{n} = 2$	4×10^{-5}	0.0021	0.076	0.28
Coherent, $\bar{n} = 8$	1×10^{-5}	0.0005	0.020	0.086
Coherent, $\bar{n} = 32$	$< 10^{-5}$	0.0001	0.005	0.023

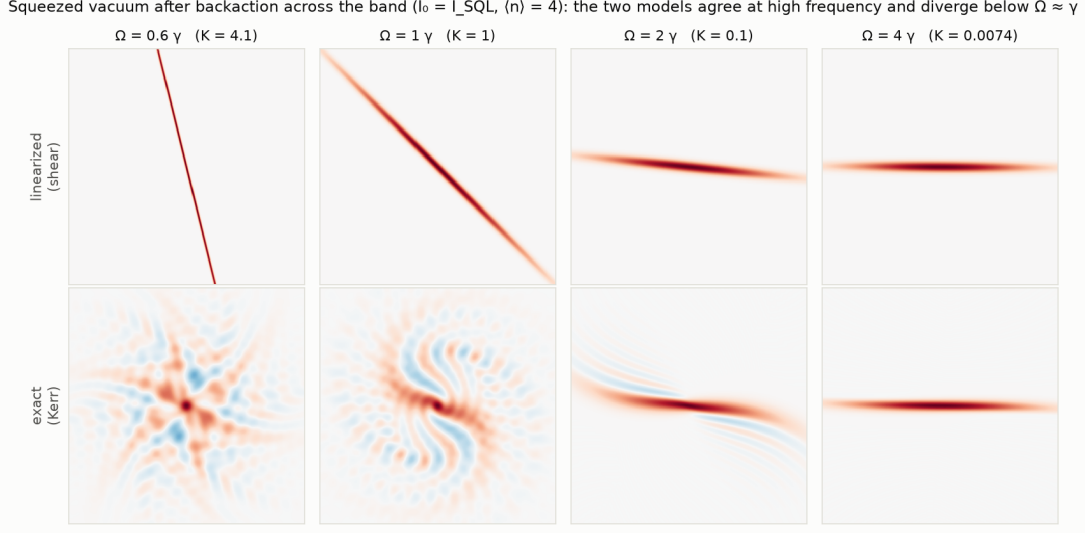


Figure 2: Squeezed vacuum after backaction at four sideband frequencies ($I_0 = I_{\text{SQL}}$, $\bar{n} = 4$): linearized shear (top row, rendered analytically from its covariance — the linearized output is Gaussian by definition) versus exact Kerr (bottom row). At $\Omega = 4\gamma$ ($\mathcal{K} \approx 0.007$) the rows are indistinguishable; at 2γ the exact state develops its first non-Gaussian curl; at and below $\Omega = \gamma$ the exact interaction wraps the state into interference-rich structures while the linearized model continues to predict a straight, thinner line.

4 Interpretation and caveats

The linearized description is a mean-field expansion about a c-number carrier; its small parameter is the ratio of photon-number-fluctuation scale to carrier amplitude. Coherent states make that ratio $1/\sqrt{\bar{n}}$; Fock, cat, and squeezed-vacuum states make it $\mathcal{O}(1)$ — not because any carrier decoheres, but because these states have no single mean field to expand around.

Two caveats bound what this result does and does not show. First, it is a *single-mode* statement: in a real interferometer the carrier is a separate, macroscopically populated coherent mode, and the exotic state is injected into the sideband (dark-port) modes, so the expansion parameter involves the injected state’s moments against the *laser* amplitude, which is enormous. Our model conflates carrier and sideband; the honest multi-mode versions of the breakdown are (i) nonlinear force terms suppressed by the carrier but weighted by the injected state’s number spread, and (ii) the $\hat{n}^2$ interaction coupling sideband *pairs* at different frequencies — which is precisely the mechanism by which per-frequency independence and linearity fail together for non-Gaussian inputs. Second, as noted above, the covariance-level metric is charitable, so the quoted breakdown couplings are lower bounds.

5 What we should do

1. **Keep both interaction forms in the simulation harness.** Use the shear for frequency-resolved, physically calibrated statements ($\chi = \mathcal{K}(\Omega)$, SQL curves, comparison against the Gaussian covariance rail) and the Kerr for any conclusion about non-Gaussian states under backaction. State-comparison results should be reported under both, with disagreement

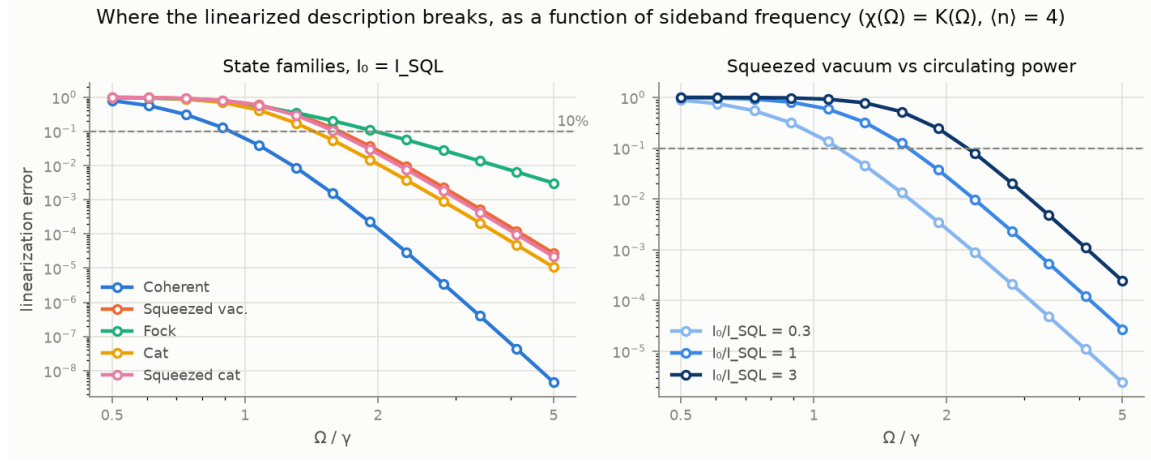


Figure 3: Linearization error versus sideband frequency (log-log; dashed guide at 10%). Left: state families at $\bar{n} = 4$, $I_0 = I_{\text{SQL}}$. The coherent carrier stays below 10% down to $\Omega \approx 0.9\gamma$, whereas the non-Gaussian families cross it at 1.5 – 2γ (Fock remains above 10^{-2} even at 5γ). Right: the same error for squeezed vacuum at three circulating powers — raising the power moves the breakdown frequency up, roughly following the contour $\mathcal{K}(\Omega) = \text{const}$.

itself reported as physics.

2. **Build the two-mode carrier-sideband model next.** An explicit coherent carrier mode tensored with a quantum sideband mode carrying the exotic state, coupled through the expanded $\hat{n}^2$ interaction, separates the artifact (“the state had to be its own carrier”) from the genuine nonlinear corrections, and gives the breakdown scale as a function of carrier power at fixed injected state.
3. **Test per-frequency independence directly.** A joint simulation of 2–3 frequency modes with the pairwise couplings induced by $\hat{n}^2$, versus the product of single-frequency channels, is the multi-mode version of this study and the direct answer to where the independent-frequency description fails for non-Gaussian states. The single-mode result predicts the relevant coupling scale.
4. **Tighten the metric.** Repeat the bridge at the level of QFI for the displacement signal (and, later, state fidelity). Since the covariance metric is the charitable one, this can only strengthen the conclusion, and it connects the breakdown directly to sensing performance.
5. **Port the classical Fisher information (homodyne) tooling to Python.** Backaction trade-offs of the standard-quantum-limit type live in fixed-measurement CFI, not in QFI (which is provably blind to the shear for phase-quadrature signals). This is the prerequisite for honest SQL statements in either model.
6. **Calibrate the regime.** Map χ_{eff} to physical interferometer parameters (circulating power, mass, measurement time) to locate where a realistic device sits on Fig. 3 for the sideband powers at which optimized non-Gaussian states would actually be injected.

Reproducibility. All numbers and figures come from the self-contained `backaction-study/`

folder (branch `claude/quantum-sensing-backaction-jn5ty7`): `run_study.py` (experiments E5 and companions; `run_bridge_frequency()` for the figures shown here), `results/bridge_results.csv`, and the regression test `test_kerr_linearizes_to_shear`, which pins the bridge Eq. (4) and its $1/\bar{n}$ convergence. Abstract-coupling (χ_{eff} -axis) versions of all three figures are also generated (`results/fig-bridge*.png`).